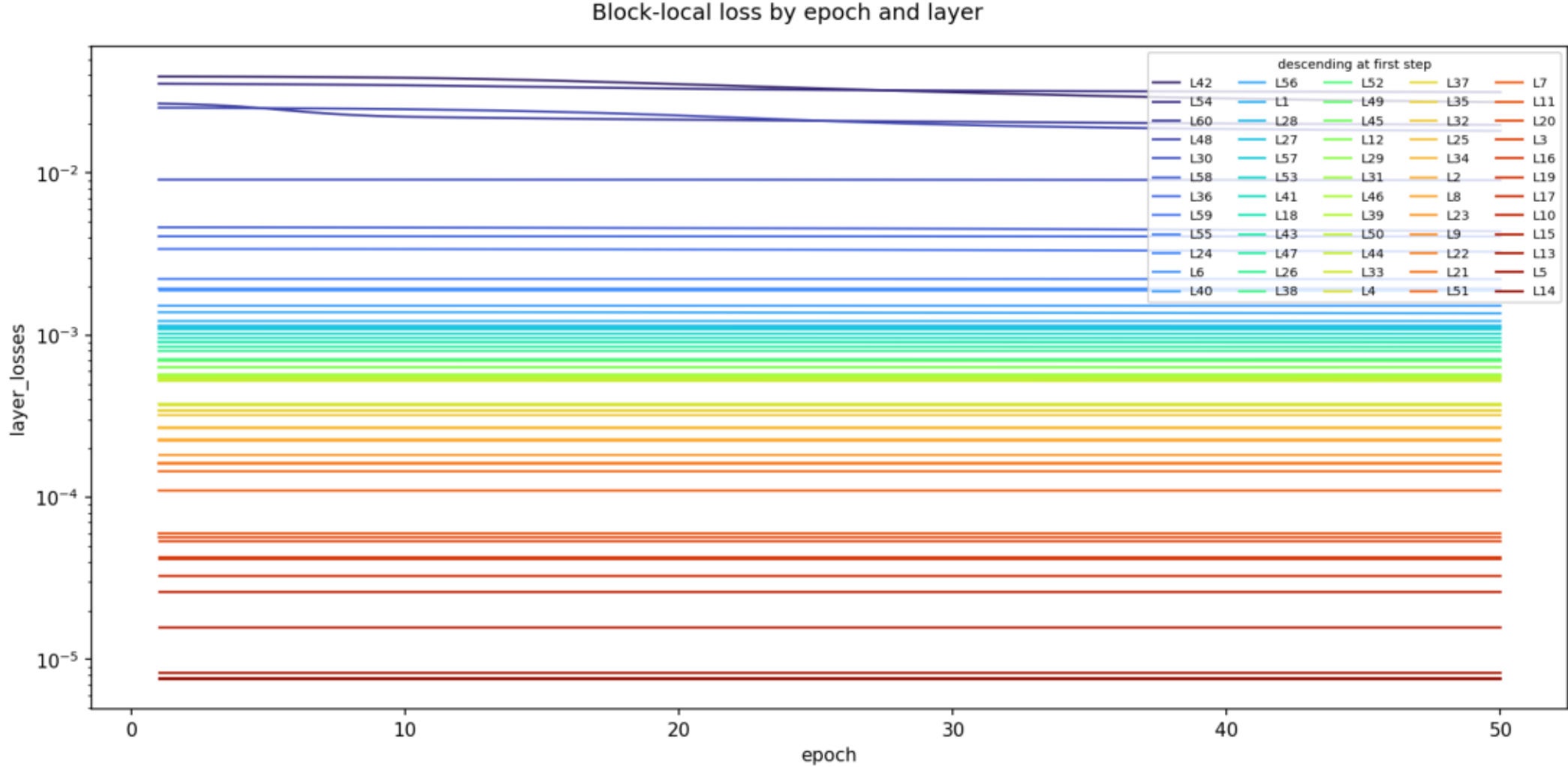


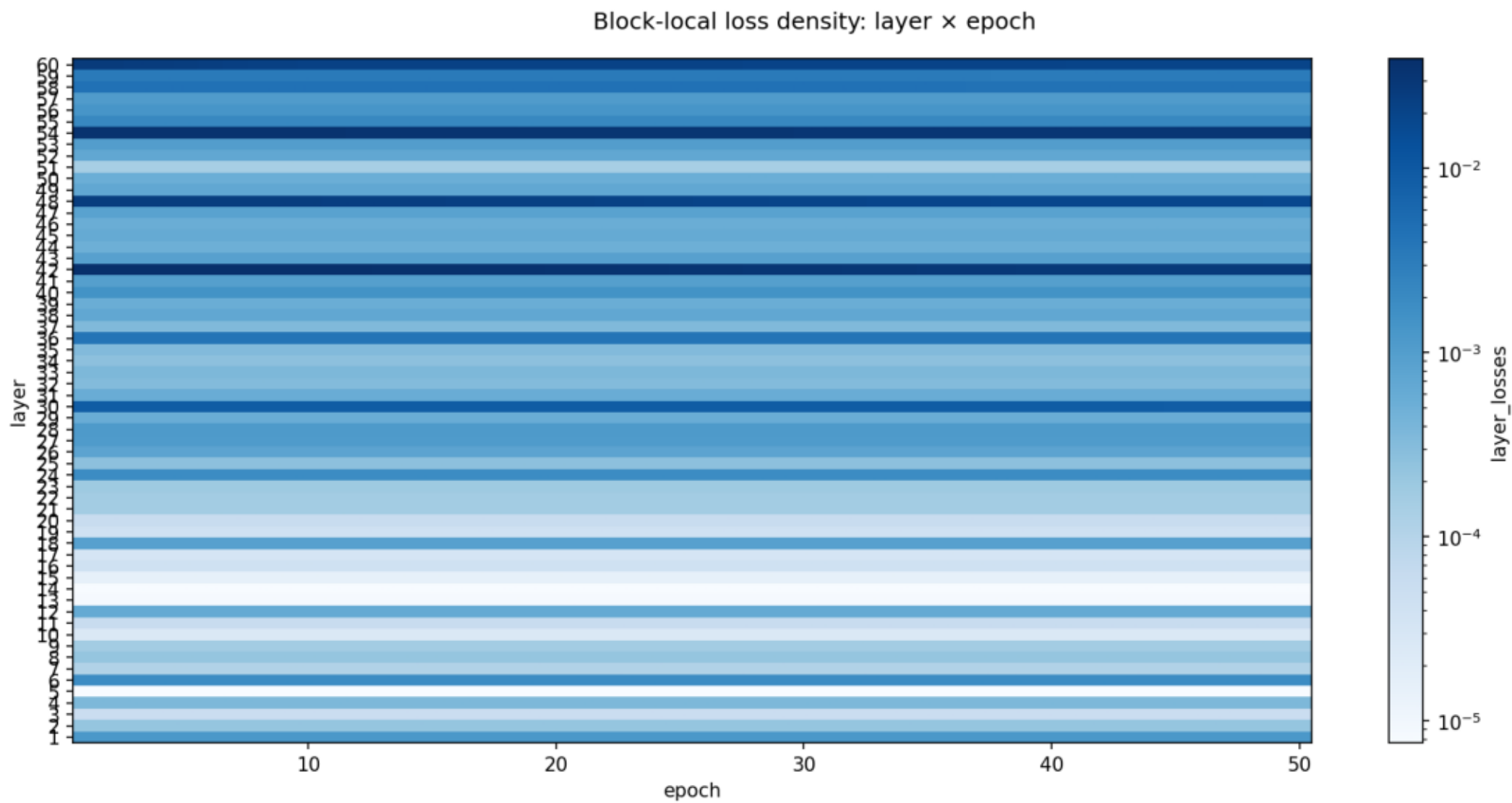
Pipeline-v4.6 report — campaign50_g31b_dense_ppp4

Fundamental datum	Value
Run	campaign50_g31b_dense_ppp4
Model identity	google/gemma-4-31B-it
Student initialization	google/gemma-4-31B-it
Source commit	ab666f0a855a
Launch id	v4-20260725134500-3362578
Runtime dirty	False
Pipeline	v4.6 / 4 stages / splits [15, 30, 45]
Training objective	huber
Loss kind	huber
LoRA	rank 16; alpha 32; packed experts False
Micro-batch / cohorts	8 / 259 (width 7-8)
Dataset	2071 items; sha 575b9dea35e0
Epochs represented	0-50
Latest stage epoch time	29.53-34.26 s
Latest CE / KL evaluation loss	10.212945377740471 / 10.204102038018156
Latest student argmax / exact-sequence	0.46806242862580894 / 0.001931434089811685
Latest recall word accuracy	0.03896664139330545
Latest standard macro accuracy	0.2916666666666667
Evaluation coverage	2071 items / 78810 tokens

Block-local loss by epoch and layer

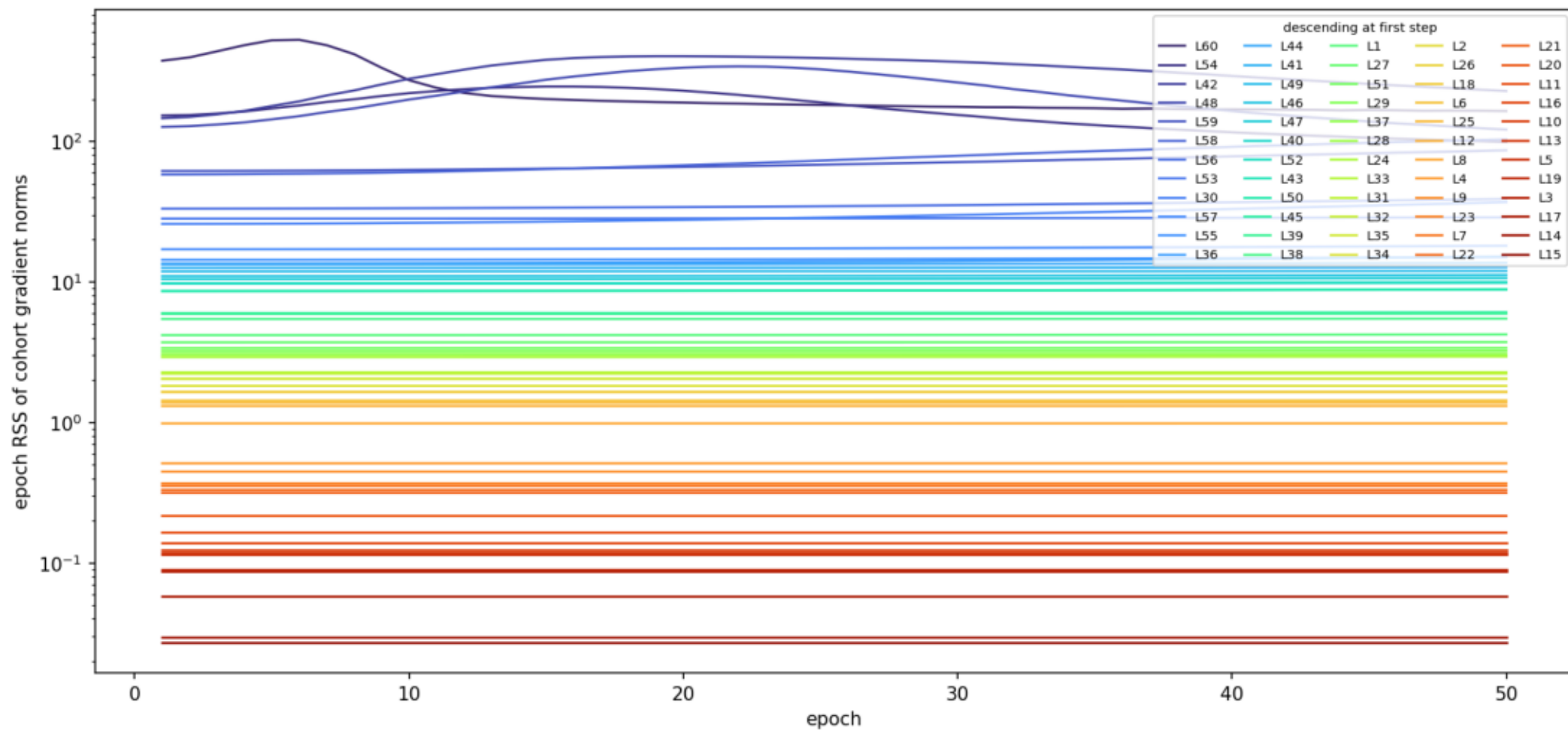


Block-local loss density: layer × epoch



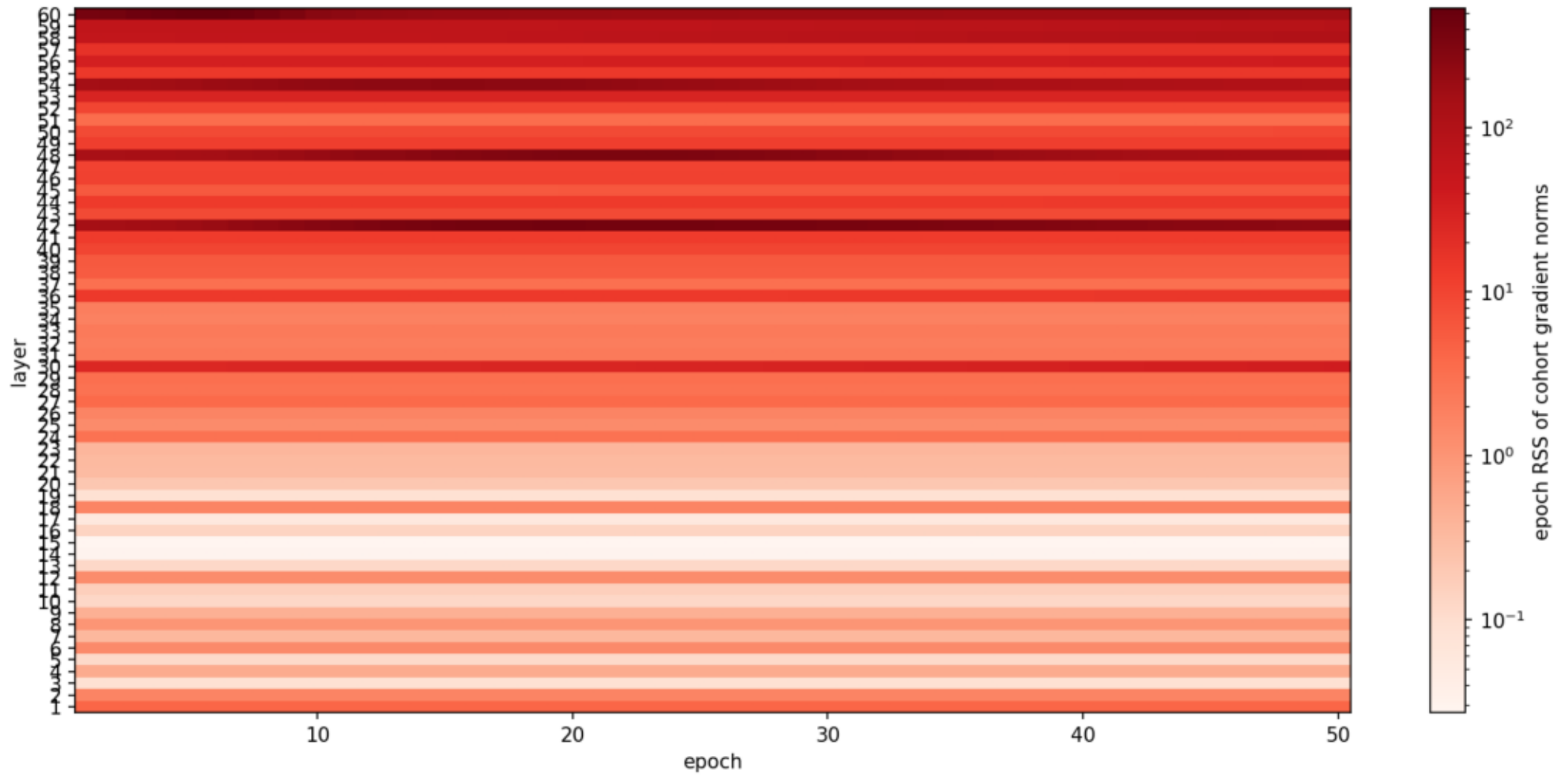
Epoch root-sum-square of cohort gradient norms by layer

Epoch root-sum-square of cohort gradient norms by layer



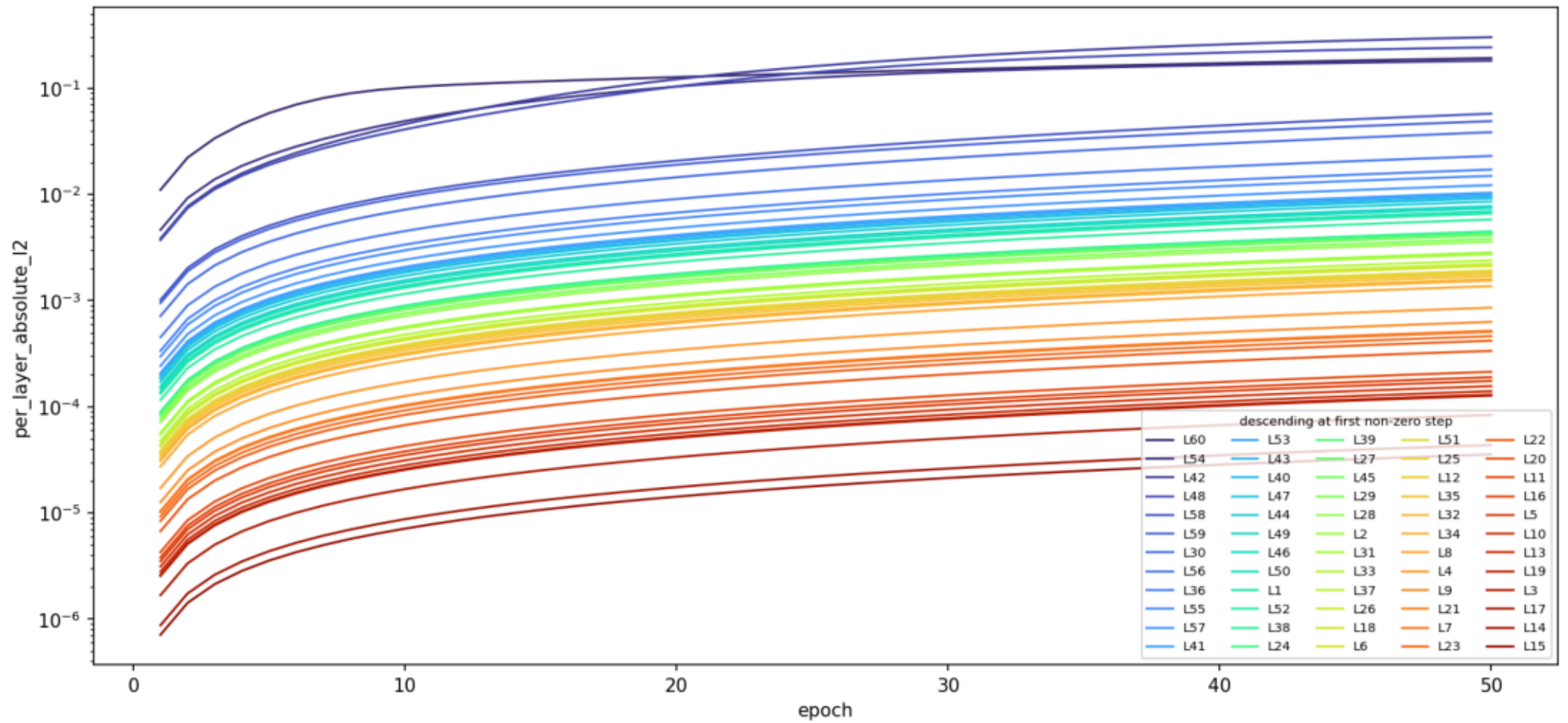
Epoch gradient RSS density: layer × epoch

Epoch gradient RSS density: layer × epoch



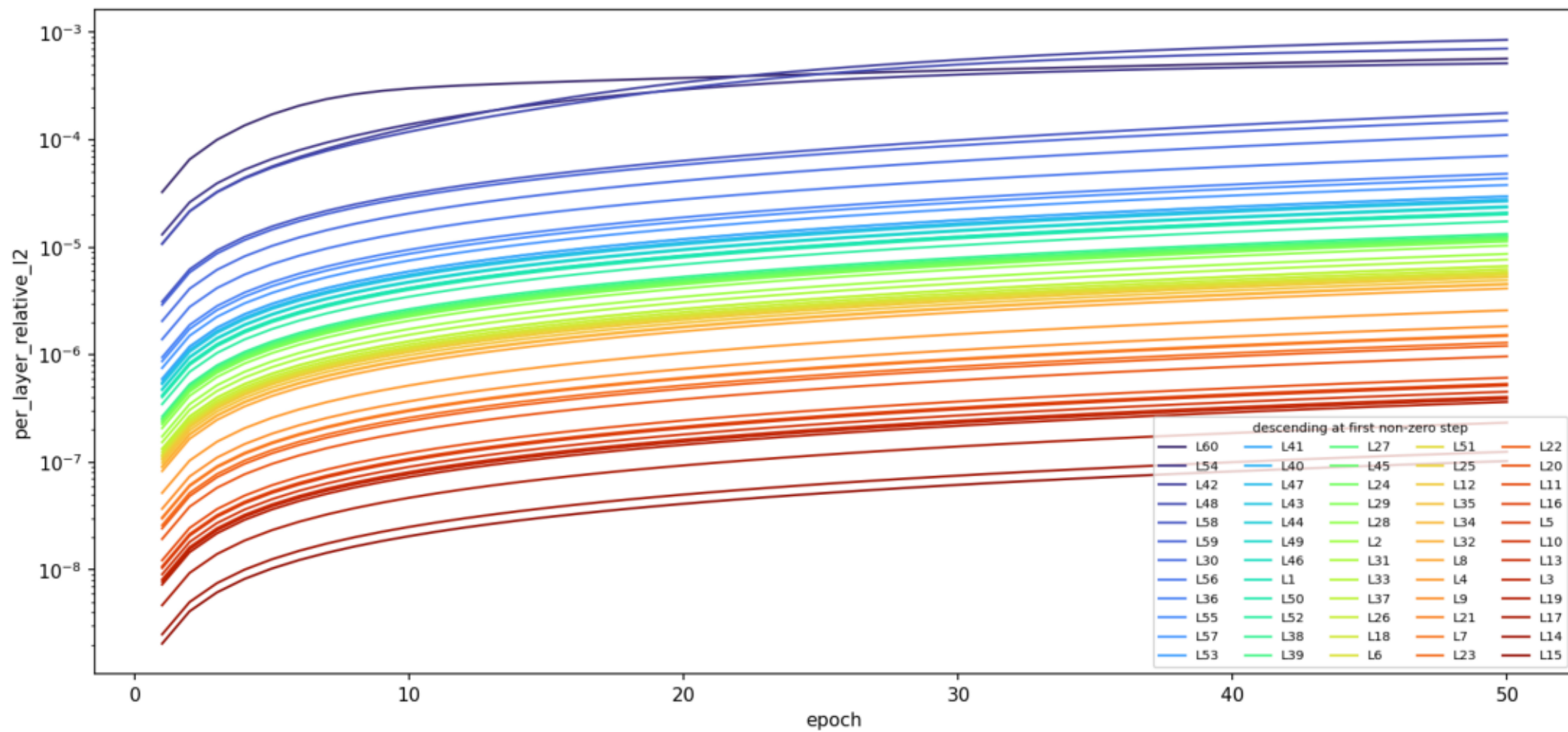
Absolute effective LoRA weight delta

Absolute effective LoRA weight delta



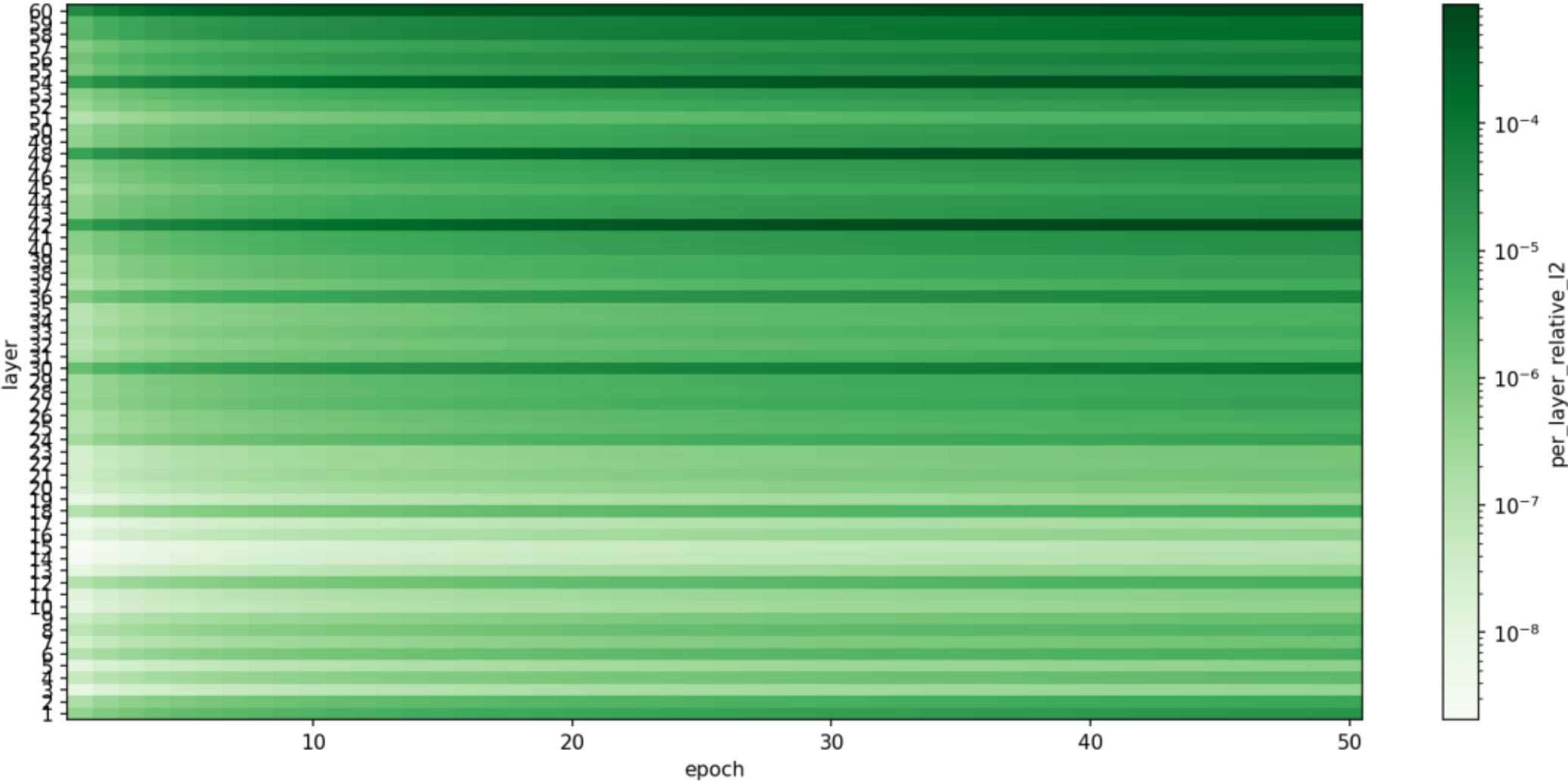
Relative effective LoRA weight delta

Relative effective LoRA weight delta



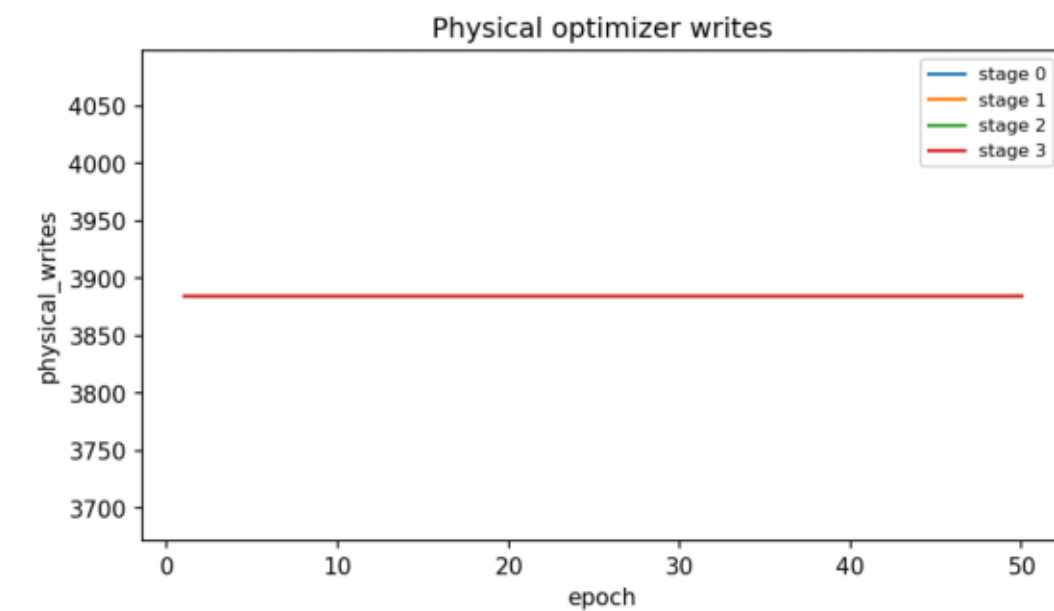
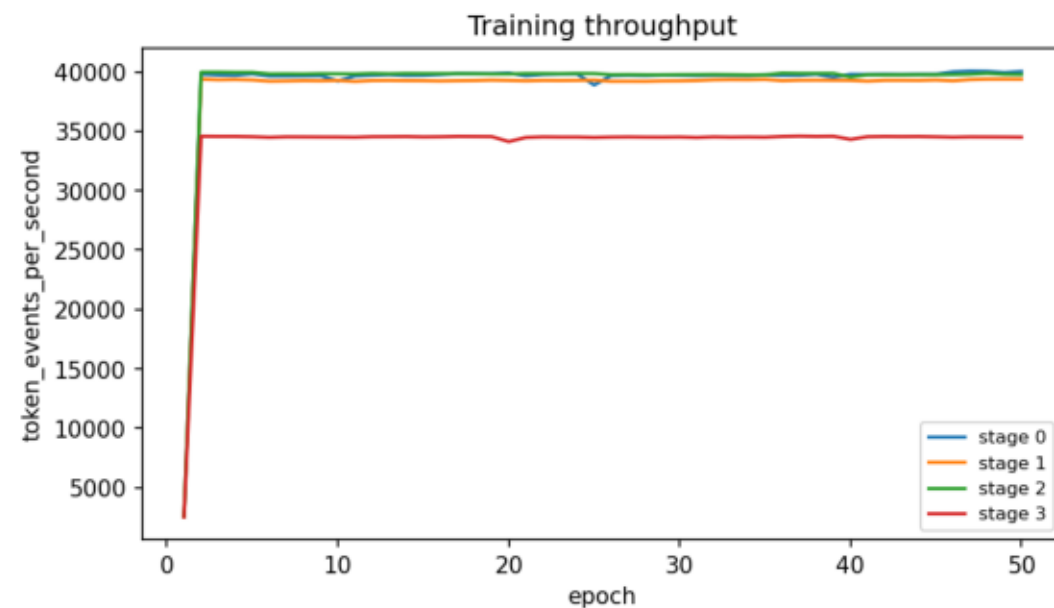
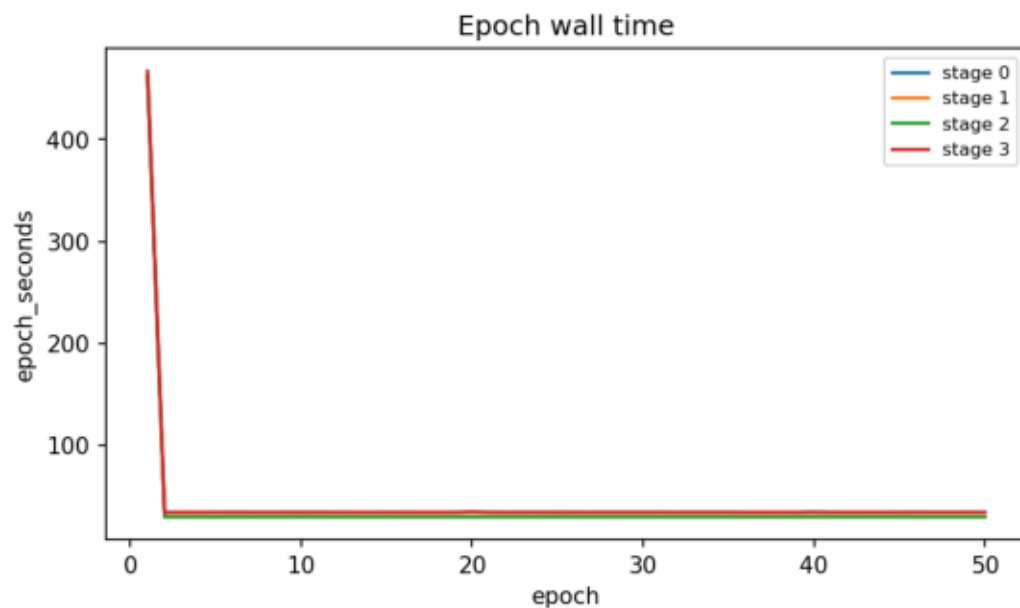
Relative effective LoRA weight-delta density: layer × epoch

Relative effective LoRA weight-delta density: layer × epoch

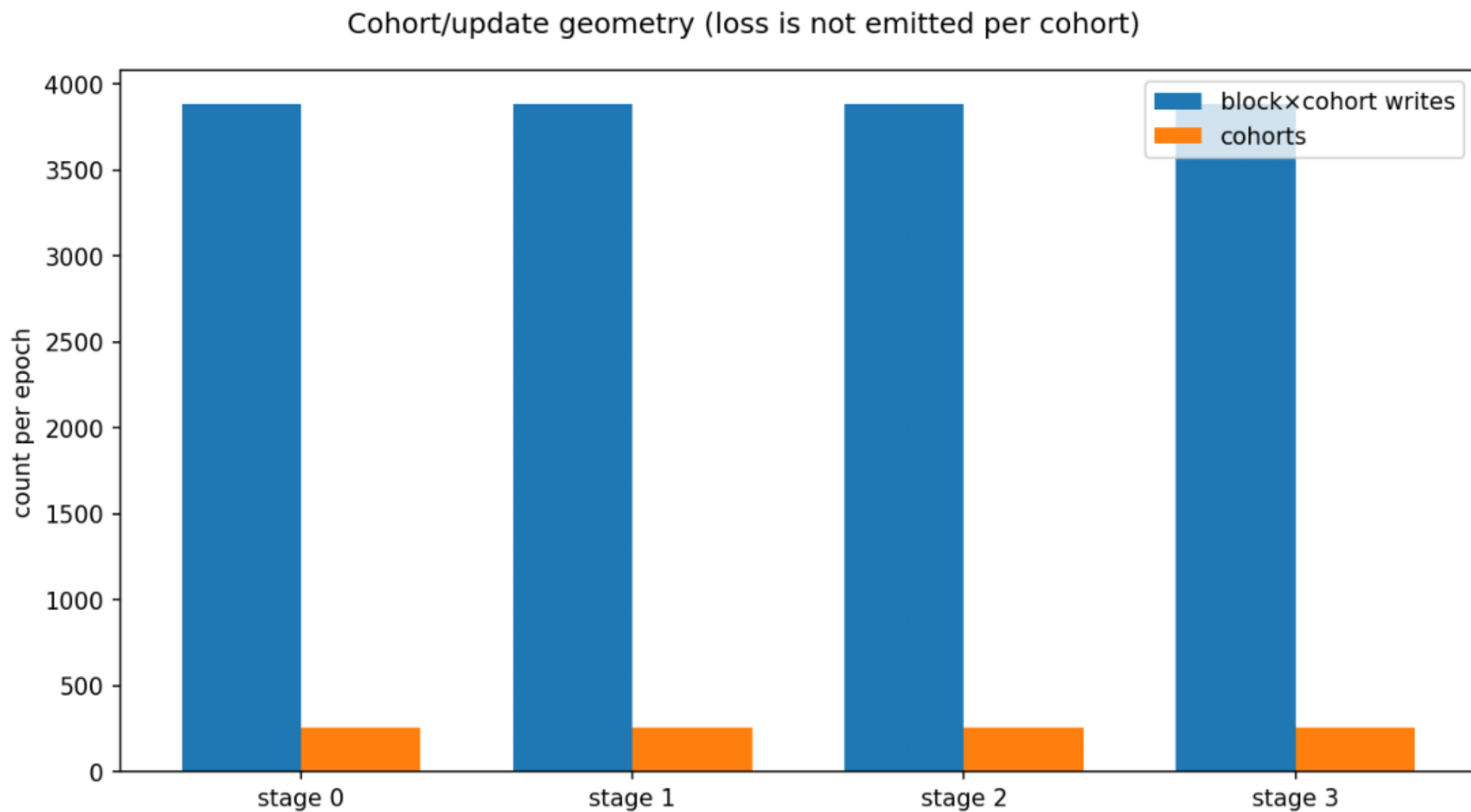


Runtime and update geometry by epoch

Runtime and update geometry by epoch

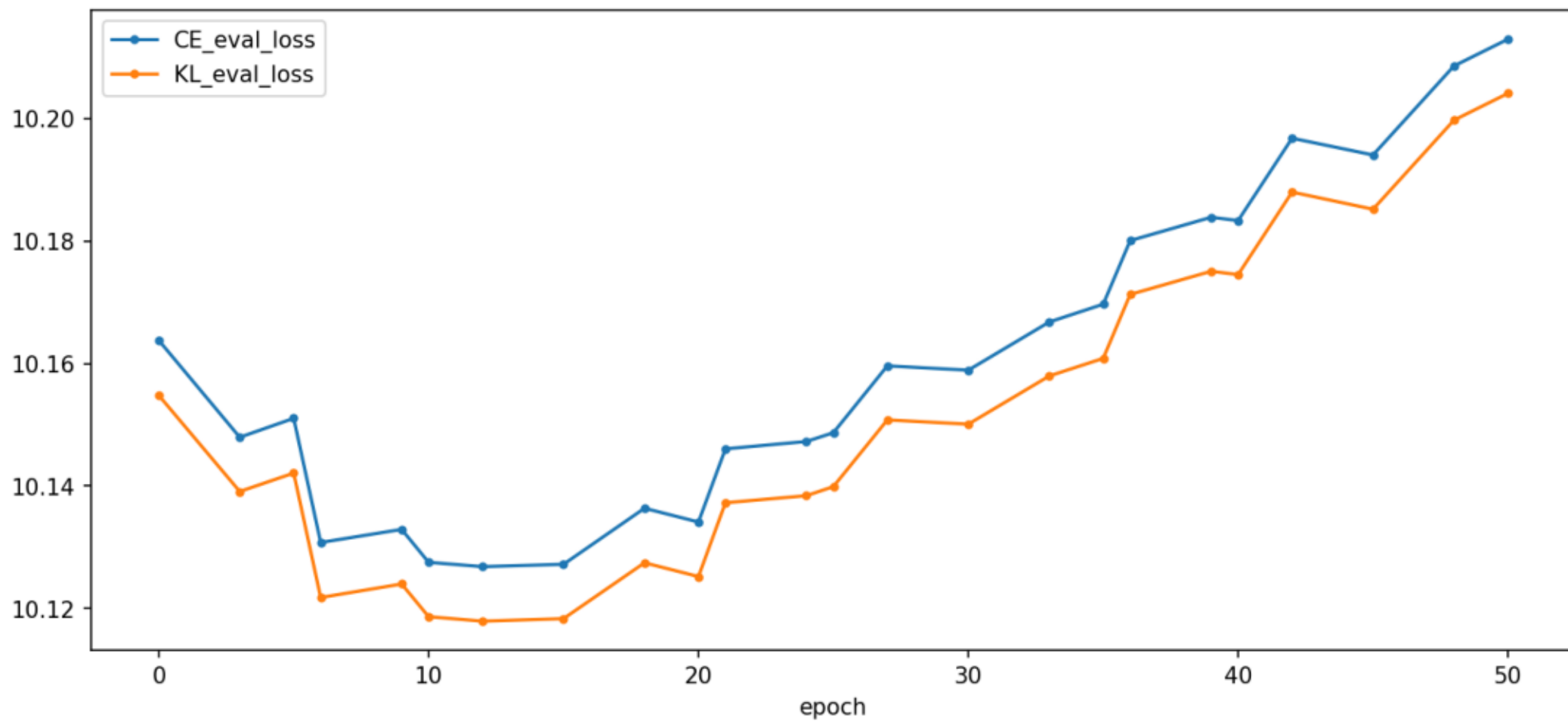


Cohort/update geometry (loss is not emitted per cohort)



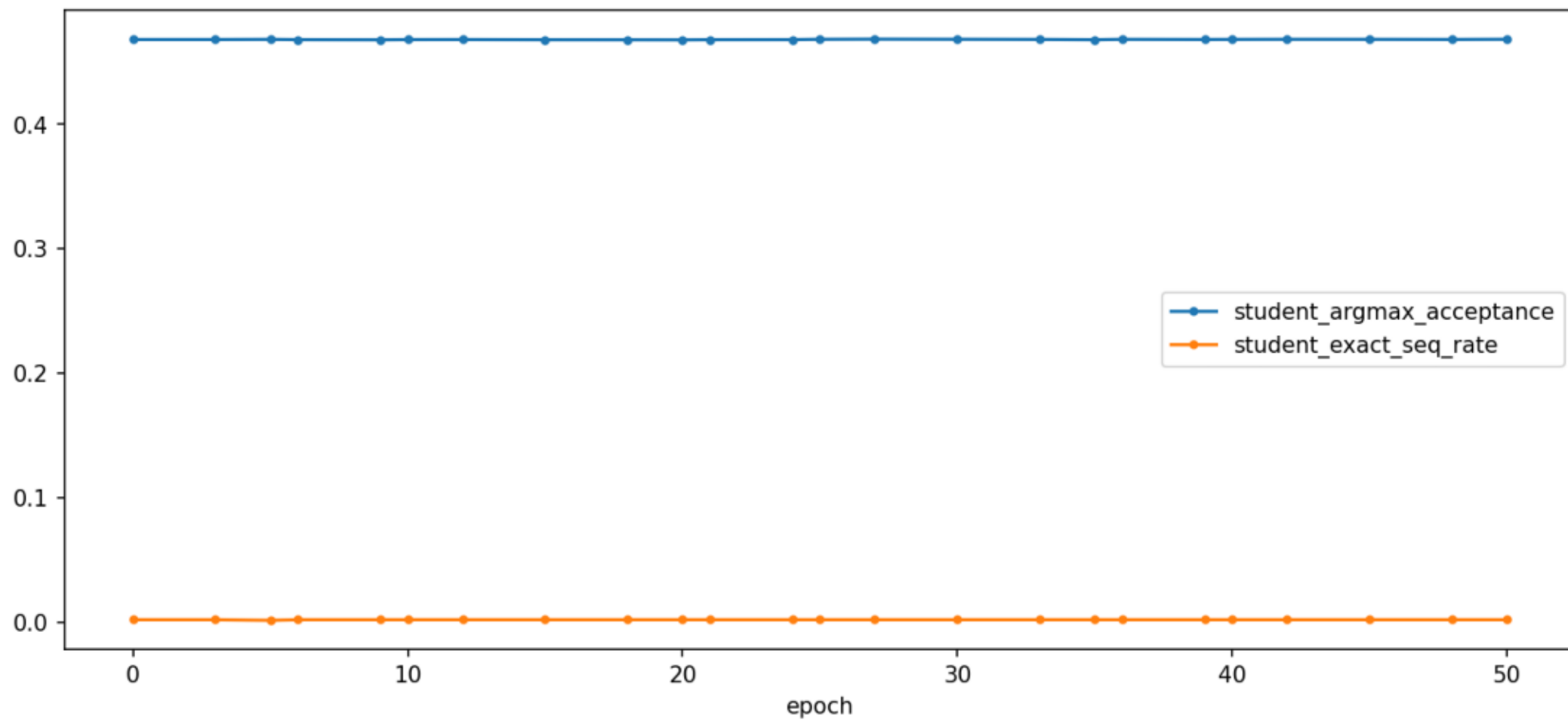
Whole-corpus evaluation losses

Whole-corpus evaluation losses

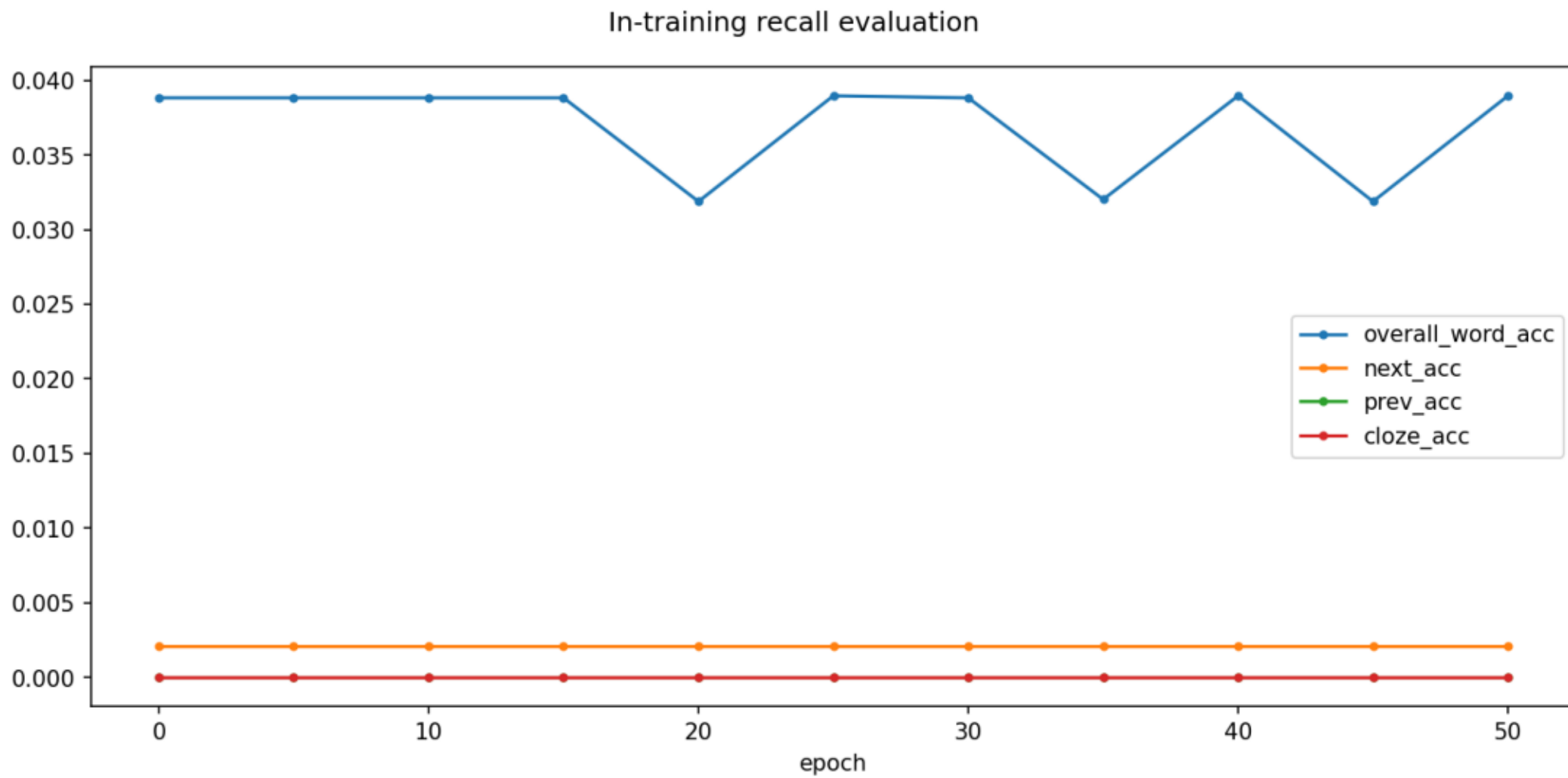


Student evaluation acceptance

Student evaluation acceptance

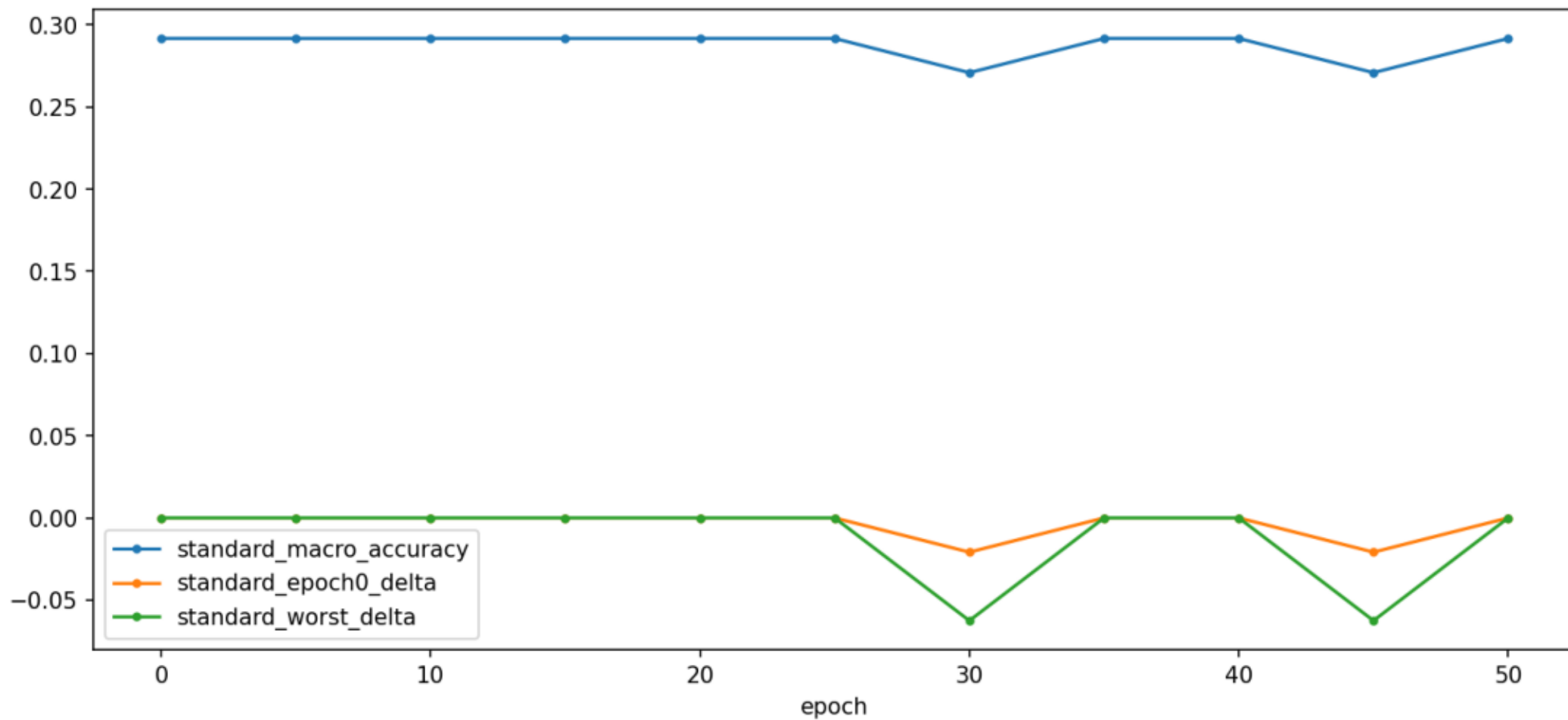


In-training recall evaluation



Standard benchmark evaluation

Standard benchmark evaluation



Evaluation runtime components

